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Apptronik and TI

Building innovative humanoids for the future

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Summary

In 2023, Apptronik announced upgrades to their humanoid robot, Apollo. As part of that development, Apptronik pulled TI in as a collaborator to support its robot's need for safe and efficient movement. Apollo utilizes Apptronik's custom linear and rotary actuators, which will be improved by TI's functional safety compliance, motor control and power management. Together, Apptronik and TI can meet their goal of enabling a dexterous humanoid robot that can operate safely and effectively in close proximity with people.

"We're thrilled to collaborate with TI as part of our commitment to building safe humanoid robots. Texas Instruments' longstanding dedication to safety in product design is a crucial component in enabling humanoids to work seamlessly alongside people," said Parker Conroy, director of product at Apptronik.

Challenge

In order to achieve human-like movement and dexterity, a humanoid robot needs a significant amount of servo drives and electronic complexity in a very small form factor. Because humanoid robots must operate reliably around humans and other robots, and adhere to evolving safety requirements for factory and warehouse automation, Apptronik needed to include safety-enabling technologies in its design.

Solution

TI's portfolio of motor-control products, with various power and size configurations addressed Apollo's needs, allowing the robots' motors to work together in unison and maintain precise and efficient movement. By working closely with TI on hardware component selection and functional safety methodologies, Apollo could also meet current and future safety requirements in a fast growing and evolving robot type.

Results

Apptronik's first commercial humanoid robot fulfills its mission of building safe, AI-powered humanoid robots that are designed for friendly interaction with workers, mass manufacturability, general-purpose capability, functional performance and safety.

Apptronik develops safe and versatile humanoids with TI expertise

Humanoids are emerging as a versatile and viable way to support industries such as manufacturing and logistics. These robots resemble humans in shape and function; they're equipped with sensors, actuators (mechanical "muscles") and advanced software that enable them to recognize objects and people; interact with their environment; and perform tasks such as walking, lifting, bending, rotating and gripping.

With TI motor-control technology, Apollo can execute accurate, precise and efficient movements coordinated over numerous motors and joints. This enables humanoid robots to walk on two legs with balance and stability, or to perform manipulative tasks such as grasping objects of different sizes and shapes.

Humanoids are still emerging and developing as viable solutions in different industries; because of this, no official safety standard has been published for this type of robot. Nonetheless, safety is a crucial design

challenge when integrating a humanoid robot into a busy environment with humans, and Appttronik prioritizes safety and is helping to influence and maintain safety standards, specifically for humanoid robots.

“The QML Class P packaging standard enables more advanced computing in space, such as how satellites and other spacecraft can autonomously process data and make decisions in orbit as opposed to beaming data back down to Earth,” said Javier Valle, product line manager for space power at our company. “More processing capability also requires greater power. With TI’s QML Class P portfolio, we increase the efficiency of the power supply while reducing the size of the overall package, resulting in much higher power density.”

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One prominent challenge of building a successful humanoid robot is developing human-like movement and dexterity. In order to achieve this, Appttronik incorporated TI’s motor-control technologies. TI microcontrollers and gate drivers, as well as analog parts deliver precise, safe and power-efficient motor control. Since Apollo has a large number of motors working together in a compact, battery-powered application, these considerations are necessary for its function.

Apollo uses TI technology that has been functional safety-certified to Technischer Überwachungsverein (TÜV SÜD) by working closely with TI on hardware component selection and functional safety methodologies. TI provides innovative and reliable parts to help Appttronik prepare for any upcoming system-level functional safety certifications specific to humanoid robots. TI also provides component-level documentation such as functional safety failure-in-time rate and failure modes, effects and diagnostics analysis.

“In order to effectively deploy humanoids in public-facing environments, we believe it’s essential to partner with industry leaders like TI in developing robust safety methodologies for future iterations of Apollo and other related projects,” Conroy said.

Outfitted with TI motor-control solutions and safety-certified technology, Apollo can work in warehouses and factories to complete repetitive and labor-intensive activities such as each picking, moving materials, delivering totes of kitted parts, and bringing parts to staff on the production line. Humanoid robots can improve safety, increase efficiency, and give human workers an opportunity to perform higher-value tasks that are less physically demanding.

Humanoid robots are currently most visible in manufacturing and logistics, and excel in performing repetitive and/or physically demanding tasks. As Apollo gains increased dexterity and the ability to perform more complex actions, it has the potential to influence several more industries. In the long term, humanoid robots like Apollo have the potential to make a positive impact in settings such as retail, medical, elder care or the home.

“Appttronik and TI are continuing to collaborate on enhanced safety methodologies and new complex motor-control capabilities, which will continue to help strengthen the versatility of these robots,” said Conroy.

About Appttronik

Appttronik is a human-centered robotics company developing AI-powered humanoid robots. Their goal is to create human helpers to support humanity in every facet of life. Appttronik’s robot, Apollo, is designed to collaborate thoughtfully with humans—initially in critical industries such as manufacturing and logistics, with future applications in healthcare, the home, and beyond. Apollo is the culmination of nearly a decade of development, drawing on Appttronik’s extensive work on 15 previous robots, including NASA’s Valkyrie robot. Appttronik started out of the Human Centered Robotics Lab at the University of Texas at Austin and has 150

employees.

Learn more about solutions for humanoid robots

Design safer, smarter and more compact humanoid robots with our system-level solutions. By leveraging our advanced gallium nitride (GaN), gate driver and precision sensing devices in your robot designs, you can drive progress across industries to increase productivity, enhance safety, and improve daily living, all while meeting your cost and performance requirements.